

Peutz-Jeghers Syndrome

Peutz-Jeghers syndrome (PJS) is an autosomal dominant cancer predisposition syndrome first described by Peutz (1921) and Jeghers (1949). Mucocutaneous pigmentation and intestinal hamartomatous polyposis are hallmarks of the disease. The pigimentary lesions resemble those seen in Carney complex, consisting of small hyperpigmented macules that typically appear in childhood (not at birth) on the lips and buccal mucosa. These lesions may also involve the eyelids, hands, and feet.

The most common malignancies associated with PJS are gastrointestinal, including cancers of the small intestine, colorectum, stomach, and pancreas. In addition, nongastrointestinal neoplasms such as breast, cervical, and endocrine tumors (thyroid, testicular, ovarian) are frequently observed. More than half of all PJS cases are attributed to mutations in the *STK11* (also known as *LKB1*) gene, a serine/threonine kinase that functions as a tumor suppressor gene. Given the high cancer risk, close surveillance of PJS patients is recommended starting at a young age.

LEOPARD Syndrome

LEOPARD syndrome (LS) is an autosomal dominant genodermatosis. The acronym LEOPARD stands for: - **L**entigines - **E**lectrocardiogram conduction defects - **O**cular hypertelorism - **P**ulmonary stenosis - **A**bnormalities of genitalia - **R**etardation of growth - **A**nd sensorineural **D**eafness

Characteristic lentigines usually develop during infancy or early childhood. Clinical diagnosis is based on typical facial features, the presence of hypertrophic cardiomyopathy, and café-au-lait macules. LS is caused by mutations in the *PTPN11* gene, which encodes the protein tyrosine phosphatase SHP-2, located on chromosome 12. This syndrome is allelic to Noonan syndrome and shares several clinical features with it.

Other Lentiginosis Syndromes

Other conditions associated with lentiginosis include: - Carney complex - Arterial dissection and lentiginosis - Laugier-Hunziker disease - Familial benign lentiginosis - Bannayan-Ruvalcaba-Riley syndrome (BRRS) - Centrofacial lentiginosis - Segmental and agminated lentiginosis

In recent years, genetic loci and causative genes have been identified for Carney complex, PJS, and BRRS.